

Model proofs

Proof of Proposition ??. By a change of variable, rewrite:

$$\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \rightarrow \begin{pmatrix} x_1 \\ u \end{pmatrix} \text{ where } u = \frac{ax_1 + (1-a)x_2}{\sqrt{a^2s^2 + (1-a)^2S^2 + 2a(1-a)\sigma}} = \frac{ax_1 + (1-a)x_2}{\sigma_I}$$

is the attractiveness rescaled to $\mathcal{N}(0, 1)$. Thus,

$$\begin{pmatrix} x_1 \\ u \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ a/\sigma_I & (1-a)/\sigma_I \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}.$$

Note that the means are still zero, but the covariance of (x_1, u) is:

$$\begin{aligned} \mathbb{C} \begin{pmatrix} x_1 \\ u \end{pmatrix} &= \begin{pmatrix} 1 & 0 \\ a/\sigma_I & (1-a)/\sigma_I \end{pmatrix} \begin{pmatrix} s^2 & \sigma \\ \sigma & S^2 \end{pmatrix} \begin{pmatrix} 1 & a/\sigma_I \\ 0 & (1-a)/\sigma_I \end{pmatrix} \\ &= \begin{pmatrix} s^2 & A \\ A & 1 \end{pmatrix} \end{aligned}$$

where

$$A = \frac{as^2 + (1-a)\sigma}{\sqrt{a^2s^2 + (1-a)^2S^2 + 2a(1-a)\sigma}} = \frac{as^2 + (1-a)\sigma}{\sigma_I}.$$

Under SGAM, individual $\begin{pmatrix} x_1 \\ u \end{pmatrix}$ is matched with $\begin{pmatrix} y_1 \\ v \end{pmatrix}$ such that $u = v = t$.

The distribution of t is $\mathcal{N}(0, 1)$. Therefore the vector $\begin{pmatrix} x_1 \\ y_1 \\ t \end{pmatrix}$ is normally distributed,

with mean 0, and covariance

$$\Sigma = \begin{pmatrix} s^2 & A^2 & A \\ A^2 & s^2 & A \\ A & A & 1 \end{pmatrix}$$

Finally, we are interested in

$$\begin{pmatrix} x_1 \\ x_2 \\ y_1 \\ y_2 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ -\frac{a}{1-a} & 0 & \frac{\sigma_I}{1-a} \\ 0 & 1 & 0 \\ 0 & -\frac{a}{1-a} & \frac{\sigma_I}{1-a} \end{pmatrix} \begin{pmatrix} x_1 \\ y_1 \\ t \end{pmatrix}$$

therefore again the means are 0 and

$$\begin{aligned} \mathbb{C} \begin{pmatrix} x_1 \\ x_2 \\ y_1 \\ y_2 \end{pmatrix} &= \begin{pmatrix} 1 & 0 & 0 \\ -\frac{a}{1-a} & 0 & \frac{\sigma_I}{1-a} \\ 0 & 1 & 0 \\ 0 & -\frac{a}{1-a} & \frac{\sigma_I}{1-a} \end{pmatrix} \Sigma \begin{pmatrix} 1 & 0 & 0 \\ -\frac{a}{1-a} & 0 & \frac{\sigma_I}{1-a} \\ 0 & 1 & 0 \\ 0 & -\frac{a}{1-a} & \frac{\sigma_I}{1-a} \end{pmatrix}^T \\ &= \begin{pmatrix} s^2 & \sigma & A^2 & AC \\ \sigma & S^2 & AC & C^2 \\ A^2 & AC & s^2 & \sigma \\ AC & C^2 & \sigma & S^2 \end{pmatrix} \end{aligned}$$

where:

$$\begin{aligned} A &= \frac{as^2 + (1-a)\sigma}{\sigma_I} \text{ and} \\ C &= \frac{a\sigma + (1-a)S^2}{\sigma_I}. \end{aligned}$$

□

Lemma 1. $C^2 \leq S^2$, with strict inequality if $a > 0$ and $\sigma < 1$.

Proof. Write

$$\begin{aligned} C^2 &= \frac{a^2\sigma^2 + (1-a)^2S^4 + 2a(1-a)\sigma S^2}{a^2s^2 + (1-a)^2S^2 + 2a(1-a)\sigma} \\ &\leq \frac{a^2s^2S^2 + (1-a)^2S^4 + 2a(1-a)\sigma S^2}{a^2s^2 + (1-a)^2S^2 + 2a(1-a)\sigma}, \text{ since } \sigma/sS = \text{Corr}(x_1, x_2) \leq 1 \\ &= S^2 \end{aligned}$$

and observe that the inequality is strict if $a > 0$ and $\sigma < sS$.

□

Proof of Claim ??. Under random matching, the joint distribution of $(\frac{\tau}{2}(x_1 + y_1) + \varepsilon, x_2, y_2)$ is normal with mean $(0, 0, 0)$ and covariance

$$\mathbb{C} = \begin{pmatrix} \frac{\tau^2}{2}(s^2 + \sigma) + 1 & \frac{\tau}{2}\sigma & \frac{\tau}{2}\sigma \\ \frac{\tau}{2}\sigma & S^2 & 0 \\ \frac{\tau}{2}\sigma & 0 & S^2 \end{pmatrix}$$

Using the matrix formula for the conditional mean of normal variables,

$$\begin{aligned} \mathbb{E} \left[\frac{\tau}{2}(x_1 + y_1) + \varepsilon \mid x_2 = v, y_2 = w \right] &= \left(\frac{\tau}{2}\sigma \quad \frac{\tau}{2}\sigma \right) \begin{pmatrix} S^2 & 0 \\ 0 & S^2 \end{pmatrix}^{-1} \begin{pmatrix} v \\ w \end{pmatrix} \\ &= \frac{\sigma\tau}{2S^2} (v + w) \end{aligned}$$

In particular, if $\sigma = 0$, this expectation is equal to 0. $\square$

Proof of Claim ??. From (??), the joint distribution of $(\frac{\tau}{2}(x_1 + y_1) + \varepsilon, x_2, y_2)$ is normal with mean $(0, 0, 0)$ and covariance

$$\Sigma = \begin{pmatrix} \frac{1}{2}\tau^2(A^2 + s^2) + 1 & \frac{\tau}{2}(\sigma + AC) & \frac{\tau}{2}(\sigma + AC) \\ \frac{\tau}{2}(\sigma + AC) & S^2 & C^2 \\ \frac{\tau}{2}(\sigma + AC) & C^2 & S^2 \end{pmatrix}$$

Therefore

$$\begin{aligned} \mathbb{E} \left[\frac{\tau}{2}(x_1 + y_1) + \varepsilon \mid x_2 = v, y_2 = w \right] &= \left(\frac{\tau}{2}(\sigma + AC) \quad \frac{\tau}{2}(\sigma + AC) \right) \begin{pmatrix} S^2 & C^2 \\ C^2 & S^2 \end{pmatrix}^{-1} \begin{pmatrix} v \\ w \end{pmatrix} \\ &= \frac{1}{2}\tau \frac{\sigma + AC}{C^2 + S^2} (v + w) \end{aligned} \tag{1}$$

In particular, if $\sigma = 0$, we have

$$\begin{aligned} A &= \frac{as^2}{\sqrt{a^2s^2 + (1-a)^2 S^2}} \quad \text{and} \\ C &= \frac{(1-a)S^2}{\sqrt{a^2s^2 + (1-a)^2 S^2}}, \end{aligned}$$

and (1) becomes

$$\begin{aligned} \mathbb{E} \left[\frac{\tau}{2}(x_1 + y_1) + \varepsilon \mid x_2 = v, y_2 = w \right] &= \frac{1}{2}\tau \frac{a(1-a)s^2}{a^2s^2 + 2(1-a)^2 S^2} (v + w) \\ &= \frac{1}{2}\tau \frac{a(1-a)\lambda}{a^2\lambda + 2(1-a)^2} (v + w) \end{aligned}$$

where $\lambda = s^2/S^2$ is the ratio of genetic variance to wealth variance. The coefficient $\frac{a(1-a)\lambda}{a^2\lambda+2(1-a)^2}$ is increasing, then decreasing in a and is 0 for $a = 0$ or $a = 1$. $\square$

Proof of Claim ??. Under random matching, the covariance matrix for children's characteristics is:

$$\begin{aligned}\mathbb{C} &= \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} + \begin{pmatrix} \frac{\tau}{2} & 0 & \frac{\tau}{2} & 0 \\ 0 & \frac{\theta}{2} & 0 & \frac{\theta}{2} \end{pmatrix} \begin{pmatrix} s^2 & \sigma & 0 & 0 \\ \sigma & S^2 & 0 & 0 \\ 0 & 0 & s^2 & \sigma \\ 0 & 0 & \sigma & S^2 \end{pmatrix} \begin{pmatrix} \frac{1}{2}\tau & 0 \\ 0 & \frac{1}{2}\theta \\ \frac{1}{2}\tau & 0 \\ 0 & \frac{1}{2}\theta \end{pmatrix} \\ &= \begin{pmatrix} \frac{1}{2}s^2\tau^2 + 1 & \frac{1}{2}\theta\sigma\tau \\ \frac{1}{2}\theta\sigma\tau & \frac{1}{2}S^2\theta^2 + 1 \end{pmatrix}\end{aligned}$$

so that the correlation between characteristics for children is:

$$\text{Corr}(x'_1, x'_2) = \frac{\frac{1}{2}\theta\sigma\tau}{\sqrt{\frac{1}{2}\tau^2s^2 + 1}\sqrt{\frac{1}{2}\theta^2S^2 + 1}}$$

Note that $\sigma = 0$ gives a zero correlation for children as well. Also, because $\theta < 1$ and $\tau < 1$, the correlation is less than the parents' correlation of σ/sS . $\square$

Proof of Claim ??. Again applying (??), under SGAM, the correlation between children's traits is:

$$\text{Corr}(x'_1, x'_2) = \frac{\frac{1}{2}\theta\tau(\sigma + AC)}{\sqrt{\frac{1}{2}\tau^2(A^2 + s^2) + 1}\sqrt{\frac{1}{2}\theta^2(C^2 + S^2) + 1}}$$

This is positive if $\sigma = 0$ so long as $AC > 0$ i.e. $0 < a < 1$. To show it is increasing in θ , strip out constant terms and take the derivative of

$$\frac{\theta}{\sqrt{\frac{1}{2}\theta^2(C^2 + S^2) + 1}}$$

The derivative is signed by

$$\begin{aligned}&\left(\frac{1}{2}\theta^2(C^2 + S^2) + 1\right)^{0.5} - \frac{1}{2}\theta^2(C^2 + S^2) \left(\frac{1}{2}\theta^2(C^2 + S^2) + 1\right)^{-0.5} \\ &> \left(\frac{1}{2}\theta^2(C^2 + S^2) + 1\right)^{0.5} - \left(\frac{1}{2}\theta^2(C^2 + S^2) + 1\right) \left(\frac{1}{2}\theta^2(C^2 + S^2) + 1\right)^{-0.5} \\ &= 0.\end{aligned}$$

$\square$

Proof of Proposition ??. The fixed point condition on the covariance matrix is

$$\begin{pmatrix} s^2 & \sigma \\ \sigma & S^2 \end{pmatrix} = \begin{pmatrix} \frac{1}{2}s^2\tau^2 + 1 & \frac{1}{2}\theta\sigma\tau \\ \frac{1}{2}\theta\sigma\tau & \frac{1}{2}S^2\theta^2 + 1 \end{pmatrix}$$

which gives

$$s^2 = \frac{2}{2-\tau^2}, S^2 = \frac{2}{2-\theta^2}, \sigma = 0.$$

The asymptotic conditional expectation of children's genetics given parental SES is:

$$\mathbb{E} \left[\frac{\tau}{2} (x_1 + y_1) + \varepsilon \mid x_2 = v, y_2 = w \right] = 0$$

since the traits x_1, x_2, y_1, y_2 are uncorrelated.

□

Proof of Proposition ??. Start by characterizing the invariant distribution. This must satisfy:

$$\begin{pmatrix} s^2 & \sigma \\ \sigma & S^2 \end{pmatrix} = \begin{pmatrix} \frac{1}{2}\tau^2 (A^2 + s^2) + 1 & \frac{1}{2}\theta\tau (\sigma + AC) \\ \frac{1}{2}\theta\tau (\sigma + AC) & \frac{1}{2}\theta^2 (C^2 + S^2) + 1 \end{pmatrix}$$

where

$$A = \frac{as^2 + (1-a)\sigma}{\sqrt{a^2s^2 + (1-a)^2S^2 + 2a(1-a)\sigma}} \quad \text{and} \\ C = \frac{a\sigma + (1-a)S^2}{\sqrt{a^2s^2 + (1-a)^2S^2 + 2a(1-a)\sigma}},$$

Note that if the distribution converges, s^2 and S^2 must be above 1. Also, for A and C to have a real-valued solution, it must be that $a^2s^2 + (1-a)^2S^2 + 2a(1-a)\sigma > 0$; using this,

$$AC = \frac{a^2s^2\sigma + (1-a)^2S^2\sigma + a(1-a)(\sigma^2 + S^2s^2)}{a^2s^2 + (1-a)^2S^2 + 2a(1-a)\sigma} > \sigma, \text{ by } \sigma^2 < S^2s^2.$$

Since $AC > \sigma$, $\frac{\theta\tau}{2}(\sigma + AC) > \theta\tau\sigma$. If $\sigma < 0$ then $\sigma = \frac{\theta\tau}{2}(\sigma + AC) > \theta\tau\sigma > \sigma$, a contradiction. Thus $\sigma \geq 0$. Also if $\sigma = 0$ then

$$\sigma = \frac{\theta\tau}{2} \left(\frac{a(1-a)(S^2s^2)}{a^2s^2 + (1-a)^2S^2} \right)$$

which implies $a = 0$ or $a = 1$. This proves that σ is non-negative, and positive if $a \in (0, 1)$, so long as the distribution converges.

From the invariant distribution, first:

$$\sigma \left(1 - \frac{1}{2}\theta\tau\right) = \frac{\theta\tau}{2} \frac{(as^2 + (1-a)\sigma)(a\sigma + (1-a)S^2)}{a^2s^2 + (1-a)^2S^2 + 2a(1-a)\sigma}$$

or

$$\mu \left(1 - \frac{1}{2}\theta\tau\right) = \frac{\theta\tau}{2} \frac{(a\lambda + (1-a)\mu)(a\mu + (1-a))}{a^2\lambda + (1-a)^2 + 2a(1-a)\mu}$$

where

$$\lambda = s^2/S^2 \text{ and } \mu = \sigma/S^2.$$

Solving for λ gives

$$\lambda = \frac{1-a}{a} \mu \frac{(-2a + 4a\mu - 2\theta\tau + 2a\theta\tau - 3a\theta\tau\mu + 2)}{(1-a)\theta\tau - 2a\mu(1-\theta\tau)} \quad (2)$$

Then

$$\begin{aligned} s^2 \left(1 - \frac{1}{2}\tau^2\right) &= \frac{1}{2}A^2\tau^2 + 1 \\ S^2 \left(1 - \frac{1}{2}\theta^2\right) &= \frac{1}{2}C^2\theta^2 + 1 \end{aligned}$$

give

$$\begin{aligned} s^2 \left(1 - \frac{1}{2}\tau^2\right) - \frac{1}{2}A^2\tau^2 &= S^2 \left(1 - \frac{1}{2}\theta^2\right) - \frac{1}{2}C^2\theta^2, \text{ therefore} \\ \lambda \left(1 - \frac{1}{2}\tau^2\right) - \frac{1}{2}\frac{A^2}{S^2}\tau^2 &= \left(1 - \frac{1}{2}\theta^2\right) - \frac{1}{2}\frac{C^2}{S^2}\theta^2 \end{aligned}$$

Here

$$\begin{aligned} \frac{A^2}{S^2} &= \frac{(a\lambda + (1-a)\mu)^2}{D} \quad \text{and} \\ \frac{C^2}{S^2} &= \frac{(a\mu + (1-a))^2}{D}, \\ D &= a^2\lambda + (1-a)^2 + 2a(1-a)\mu \end{aligned}$$

which give a quadratic equation in μ :

$$\lambda \left(1 - \frac{1}{2}\tau^2\right) D - \frac{1}{2}(a\lambda + (1-a)\mu)^2 \tau^2 - \left(1 - \frac{1}{2}\theta^2\right) D + \frac{1}{2}(a\mu + (1-a))^2 \theta^2 = 0$$

Plugging in λ given by (2), this can be rewritten to

$$F(\mu) = \frac{(1-a+a\mu)^2}{a} \frac{N(\mu)}{D(\mu)} = 0$$

where

$$N(\mu) = X\mu^2 + Y\mu + Z, \text{ with } X, Y, Z \text{ polynomials in } a, \theta, \tau$$

and

$$D(\mu) = (\theta\tau(1-a) - 2a\mu(1-\theta\tau))^2$$

One can check that the discriminant is always positive. Therefore this has two solutions (not shown), of which only one is acceptable (it goes to the exact solution when the coefficient of μ^2 goes to 0). Writing

$$\mu = \phi_1(a, \theta, \tau)$$

for this solution:

$$\lambda = \psi(a, \theta, \tau) = \frac{1-a}{a} \phi_1(a, \theta, \tau) \frac{(-2a + 4a\phi_1(a, \theta, \tau) - 2\theta\tau + 2a\theta\tau - 3a\theta\tau\phi_1(a, \theta, \tau) + 2)}{(1-a)\theta\tau - 2a\phi_1(a, \theta, \tau)(1-\theta\tau)}.$$

Finally

$$S^2 = \frac{1}{1 - \frac{1}{2}\theta^2 - \frac{1}{2}\frac{C^2}{S^2}\theta^2} \text{ where}$$

$$\frac{C^2}{S^2} = \frac{(a\phi_1(a, \theta, \tau) + (1-a))^2}{a^2\psi(a, \theta, \tau) + (1-a)^2 + 2a(1-a)\phi_1(a, \theta, \tau)}$$

and

$$s^2 = \lambda S^2 = \frac{\psi(a, \theta, \tau)}{1 - \frac{1}{2}\theta^2 - \frac{1}{2}\frac{C^2}{S^2}\theta^2};$$

$$\sigma = \mu S^2 = \frac{\phi_1(a, \theta, \tau)}{1 - \frac{1}{2}\theta^2 - \frac{1}{2}\frac{C^2}{S^2}\theta^2}.$$

Conditional expectations of children's genetics given parents' wealth under SGAM are calculated using the same formula as before, plugging in moments of the asymptotic distribution:

$$\mathbb{E} \left[\frac{\tau}{2} (x_1 + y_1) + \varepsilon \mid x_2 = v, y_2 = w \right] = \frac{1}{2} \tau \frac{\sigma + AC}{C^2 + S^2} (v + w)$$

□

Proof of Claim ??. Since men and women have different distributions of attractiveness, we have to match them by quantiles of their respective distributions. Men's

and women's attractiveness are distributed

$$N(0, \sigma_I^2) \text{ where } \sigma_I = \sqrt{a^2 s^2 + (1-a)^2 S^2 + 2a(1-a)\sigma};$$

$$N(0, \sigma_J^2) \text{ where } \sigma_J = \sqrt{b^2 s^2 + (1-b)^2 S^2 + 2b(1-b)\sigma}.$$

Thus, men with normalized attractiveness $i(x)/\sigma_I$ match women with normalized attractiveness $j(y)/\sigma_J$.

Change variables so that

$$\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \rightarrow \begin{pmatrix} x_1 \\ u \end{pmatrix} \text{ where } u = \frac{ax_1 + (1-a)x_2}{\sigma_I};$$

$$\begin{pmatrix} y_1 \\ y_2 \end{pmatrix} \rightarrow \begin{pmatrix} y_1 \\ v \end{pmatrix} \text{ where } v = \frac{by_1 + (1-b)y_2}{\sigma_J}.$$

Thus

$$\begin{pmatrix} x_1 \\ u \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ a/\sigma_I & (1-a)/\sigma_I \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix};$$

$$\begin{pmatrix} y_1 \\ v \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ b/\sigma_J & (1-b)/\sigma_J \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \end{pmatrix}.$$

and their respective covariance matrices are

$$\begin{aligned} \mathbb{C} \begin{pmatrix} x_1 \\ u \end{pmatrix} &= \begin{pmatrix} 1 & 0 \\ a/\sigma_I & (1-a)/\sigma_I \end{pmatrix} \begin{pmatrix} s^2 & \sigma \\ \sigma & S^2 \end{pmatrix} \begin{pmatrix} 1 & a/\sigma_I \\ 0 & (1-a)/\sigma_I \end{pmatrix} \\ &= \begin{pmatrix} s^2 & A \\ A & 1 \end{pmatrix}, \text{ where } A = \frac{as^2 + (1-a)\sigma}{\sigma_I}; \end{aligned}$$

similarly

$$\mathbb{C} \begin{pmatrix} y_1 \\ v \end{pmatrix} = \begin{pmatrix} s^2 & B \\ B & 1 \end{pmatrix}, \text{ where } B = \frac{bs^2 + (1-b)\sigma}{\sigma_J}.$$

Under SGAM, couples have characteristics $\begin{pmatrix} x_1 \\ t \\ y_1 \\ t \end{pmatrix}$, where $\begin{pmatrix} x_1 \\ y_1 \\ t \end{pmatrix}$ is trivariate normal

with mean 0 and covariance matrix

$$\Sigma = \begin{pmatrix} s^2 & AB & A \\ AB & s^2 & B \\ A & B & 1 \end{pmatrix}$$

Lastly, we calculate the covariance matrix of couples' original characteristics. Since

$$\begin{pmatrix} x_1 \\ x_2 \\ y_1 \\ y_2 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ \frac{-a}{1-a} & 0 & \frac{\sigma_I}{1-a} \\ 0 & 1 & 0 \\ 0 & \frac{-b}{1-b} & \frac{\sigma_J}{1-b} \end{pmatrix} \begin{pmatrix} x_1 \\ y_1 \\ t \end{pmatrix}$$

we have that the mean is again 0 and the covariance matrix is

$$\begin{aligned} \mathbb{C} \begin{pmatrix} x_1 \\ x_2 \\ y_1 \\ y_2 \end{pmatrix} &= \begin{pmatrix} 1 & 0 & 0 \\ \frac{-a}{1-a} & 0 & \frac{\sigma_I}{1-a} \\ 0 & 1 & 0 \\ 0 & \frac{-b}{1-b} & \frac{\sigma_J}{1-b} \end{pmatrix} \Sigma \begin{pmatrix} 1 & \frac{-a}{1-a} & 0 & 0 \\ 0 & 0 & 1 & \frac{-b}{1-b} \\ 0 & \frac{\sigma_I}{1-a} & 0 & \frac{\sigma_J}{1-b} \end{pmatrix} \\ &= \begin{pmatrix} s^2 & \sigma & AB & AD \\ \sigma & S^2 & BC & CD \\ AB & BC & s^2 & \sigma \\ AD & CD & \sigma & S^2 \end{pmatrix} \end{aligned}$$

where

$$C = \frac{a\sigma + (1-a)S^2}{\sigma_I}; D = \frac{b\sigma + (1-b)S^2}{\sigma_J}.$$

From the above and (??) we can calculate the covariance matrix of children's characteristics as

$$\begin{aligned} \mathbb{C} \begin{pmatrix} x'_1 \\ x'_2 \end{pmatrix} &= \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} + \begin{pmatrix} \frac{\tau}{2} & 0 & \frac{\tau}{2} & 0 \\ 0 & \frac{\theta}{2} & 0 & \frac{\theta}{2} \end{pmatrix} \begin{pmatrix} s^2 & \sigma & AB & AD \\ \sigma & S^2 & BC & CD \\ AB & BC & s^2 & \sigma \\ AD & CD & \sigma & S^2 \end{pmatrix} \begin{pmatrix} \frac{\tau}{2} & 0 \\ 0 & \frac{\theta}{2} \\ \frac{\tau}{2} & 0 \\ 0 & \frac{\theta}{2} \end{pmatrix} \\ &= \begin{pmatrix} \frac{\tau^2}{2}(s^2 + AB) + 1 & \frac{\tau\theta}{4}(2\sigma + AD + BC) \\ \frac{\tau\theta}{4}(2\sigma + AD + BC) & \frac{\theta^2}{2}(S^2 + CD) + 1 \end{pmatrix}. \end{aligned}$$

Thus x'_1 and x'_2 will be positively correlated if $2\sigma + AD + BC > 0$. This is always positive if $\sigma > 0$; if $\sigma = 0$ it reduces to

$$\frac{(a + b - 2ab)s^2 S^2}{\sigma_I \sigma_J}$$

which is positive unless $a = b = 0$ or $a = b = 1$. The correlation is

$$\frac{\frac{\tau\theta}{4}(2\sigma + AD + BC)}{\sqrt{\frac{\tau^2}{2}(s^2 + AB) + 1} \sqrt{\frac{\theta^2}{2}(S^2 + CD) + 1}}$$

and taking the derivative shows it is increasing in θ , as in the proof for Claim ?? $\square$

Proof of Proposition ??. Write

$$\begin{aligned} x'_1 &= \tau \frac{x_1 + y_1}{2} + \varepsilon \\ x'_2 &= \gamma x'_1 + \theta \frac{x_2 + y_2}{2} + \eta \\ &= \gamma \tau \frac{x_1 + y_1}{2} + \theta \frac{x_2 + y_2}{2} + \eta + \gamma \varepsilon \end{aligned}$$

Since

$$\begin{pmatrix} \tau \frac{x_1 + y_1}{2} \\ \gamma \tau \frac{x_1 + y_1}{2} + \theta \frac{x_2 + y_2}{2} \end{pmatrix} = \begin{pmatrix} \frac{\tau}{2} & 0 & \frac{\tau}{2} & 0 \\ \frac{\gamma\tau}{2} & \theta & \frac{\gamma\tau}{2} & \theta \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ y_1 \\ y_2 \end{pmatrix}$$

we can use (??) to derive the covariance matrix for children:

$$\begin{aligned} \mathbb{C} &= \begin{pmatrix} 1 & \gamma \\ \gamma & 1 + \gamma^2 \end{pmatrix} + \begin{pmatrix} \frac{\tau}{2} & 0 & \frac{\tau}{2} & 0 \\ \frac{\gamma\tau}{2} & \theta & \frac{\gamma\tau}{2} & \theta \end{pmatrix} \begin{pmatrix} s^2 & \sigma & A^2 & AC \\ \sigma & S^2 & AC & C^2 \\ A^2 & AC & s^2 & \sigma \\ AC & C^2 & \sigma & S^2 \end{pmatrix} \begin{pmatrix} \frac{\tau}{2} & \frac{\gamma\tau}{2} \\ 0 & \frac{\theta}{2} \\ \frac{\tau}{2} & \frac{\gamma\tau}{2} \\ 0 & \frac{\theta}{2} \end{pmatrix} \\ &= \begin{pmatrix} \frac{\tau^2}{2}(s^2 + A^2) + 1 & \frac{\gamma\tau^2}{2}(s^2 + A^2) + \frac{\tau\theta}{2}(\sigma + AC) + \gamma \\ \frac{\gamma\tau^2}{2}(s^2 + A^2) + \frac{\tau\theta}{2}(\sigma + AC) + \gamma & \frac{\gamma^2\tau^2}{2}(s^2 + A^2) + \gamma\tau\theta(\sigma + AC) + \frac{\theta^2}{2}(S^2 + C^2) + 1 + \gamma^2 \end{pmatrix} \end{aligned}$$

The first claim in the proof follows from the covariance:

$$\frac{\gamma\tau^2}{2}(s^2 + A^2) + \frac{\tau\theta}{2}(\sigma + AC) + \gamma$$

This is increasing in γ , and positive if any of $\sigma > 0$, $\gamma > 0$, or $AC > 0$ (which holds if $0 < a < 1$ when $\sigma = 0$).

The correlation $Cov(x'_1, x'_2) / \sqrt{Var(x'_1)Var(x'_2)}$ is proportional to

$$\frac{\gamma p + q}{\sqrt{\gamma^2 p + \gamma^2 q + r}}$$

where

$$\begin{aligned} p &= \tau^2(s^2 + A^2) + 2; \\ q &= \tau\theta(\sigma + AC); \\ r &= \theta^2(S^2 + C^2) + 2. \end{aligned}$$

The derivative of this with respect to γ is signed by $pr - q^2$, which equals

$$\begin{aligned} & [\tau^2(s^2 + A^2) + 2][\theta^2(S^2 + C^2) + 2] - [\tau\theta(\sigma + AC)]^2 \\ &= \tau^2\theta^2(s^2S^2 + A^2S^2 + s^2C^2 - 2\sigma AC - \sigma^2) + 2[\theta^2(S^2 + C^2) + \tau^2(s^2 + A^2)] + 4 \end{aligned}$$

The last two terms are positive. In the first term, $\sigma^2 < s^2S^2$, and

$$\begin{aligned} 0 &< (AS - Cs)^2 \\ &= A^2S^2 + C^2s^2 - 2ACs\sigma \\ &< A^2S^2 + C^2s^2 - 2\sigma AC, \text{ again using } \sigma < sS. \end{aligned}$$

Hence the whole sum is positive.

Now we can calculate

$$\mathbb{E}[x'_2|x_2 + y_2]$$

using

$$\begin{pmatrix} \gamma\tau\frac{x_1+y_1}{2} + \theta\frac{x_2+y_2}{2} \\ x_2 \\ y_2 \end{pmatrix} = \begin{pmatrix} \frac{\gamma\tau}{2}0\frac{\theta}{2}1\frac{\gamma\tau}{2}0\frac{\theta}{2}0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ y_1 \\ y_2 \end{pmatrix}$$

to give

$$\begin{aligned} \mathbb{C} \begin{pmatrix} x'_2 \\ x_2 \\ y_2 \end{pmatrix} &= \begin{pmatrix} \frac{\gamma\tau}{2} & \frac{\theta}{2} & \frac{\gamma\tau}{2} & \frac{\theta}{2} \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} s^2 & \sigma & A^2 & AC \\ \sigma & S^2 & AC & C^2 \\ A^2 & AC & s^2 & \sigma \\ AC & C^2 & \sigma & S^2 \end{pmatrix} \begin{pmatrix} \frac{\gamma\tau}{2}00 \\ \frac{\theta}{2}10 \\ \frac{\gamma\tau}{2}00 \\ \frac{\theta}{2}01 \end{pmatrix} + \begin{pmatrix} 1 + \gamma^2 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \\ &= \begin{pmatrix} \dots & \frac{\gamma\tau}{2}(\sigma + AC) + \frac{\theta}{2}(S^2 + C^2) & \frac{\gamma\tau}{2}(\sigma + AC) + \frac{\theta}{2}(S^2 + C^2) \\ \frac{\gamma\tau}{2}(\sigma + AC) + \frac{\theta}{2}(S^2 + C^2) & S^2 & C^2 \\ \frac{\gamma\tau}{2}(\sigma + AC) + \frac{\theta}{2}(S^2 + C^2) & C^2 & S^2 \end{pmatrix}. \end{aligned}$$

Next

$$\begin{aligned} \mathbb{E}[x'_2|x_2, y_2] &= \left(\frac{\gamma\tau}{2}(\sigma + AC) + \frac{\theta}{2}(S^2 + C^2) \quad \frac{\gamma\tau}{2}(\sigma + AC) + \frac{\theta}{2}(S^2 + C^2) \right) \begin{pmatrix} S^2 & C^2 \\ C^2 & S^2 \end{pmatrix}^{-1} \begin{pmatrix} x_2 \\ y_2 \end{pmatrix} \\ &= \left(\gamma\tau \frac{\sigma + AC}{S^2 + C^2} + \theta \right) \frac{x_2 + y_2}{2} \end{aligned}$$

So long as $0 < a < 1$ or $\sigma > 0$, the coefficient on parents' wealth is thus higher than θ .

□

Proof of Proposition ??. Without loss of generality let $Ex_1 = Ex_2 = 0$. The correlation is signed by the covariance. Write K for the set of couples in the parents' generation with typical member $k = (x, y)$. Without loss of generality let $x_1 \geq y_1$. Then, since the iso-attractiveness curves defined by f are downward-sloping, $x_2 \leq y_2$. (If $a = 1$ then $x_1 = y_1$; pick x so that $x_2 \leq y_2$.) Also, for $a \in (0, 1)$, if $x_1 > y_1$ then $x_2 < y_2$.

Since $Ex_1 = Ex_2 = 0$, the covariance among the parents' generation is

$$\int_K (x_1 x_2 + y_1 y_2) / 2 \, dk$$

Write

$$\begin{aligned} x'_1 &= \tau x_1^* + \varepsilon & \text{where } x_1^* &= (x_1 + y_1) / 2 \\ x'_2 &= \theta x_2^* + \eta & \text{where } x_2^* &= (x_2 + y_2) / 2 \end{aligned}$$

and write the children's covariance as

$$Cov(x'_1, x'_2) = Cov(\tau x_1^*, \theta x_2^*) + Cov(\tau x_1^*, \eta) + Cov(\varepsilon, \theta x_2^*) + Cov(\varepsilon, \eta).$$

By independence of the shocks, the last 3 terms are zero. So we need to show that

$$Cov(\tau x_1^*, \theta x_2^*) = \tau \theta Cov(x_1^*, x_2^*) > 0$$

Write

$$Cov(x_1^*, x_2^*) = \int_K x_1^* x_2^* \, dk$$

using that $Ex_1^* = Ex_2^* = 0$.

Take a typical parent, and write

$$\begin{aligned} x_1 x_2 &= (x_1^* - \Delta_1)(x_2^* - \Delta_2) \\ y_1 y_2 &= (x_1^* + \Delta_1)(x_2^* + \Delta_2) \end{aligned}$$

where

$$\Delta_1 = (x_1 - y_1) / 2; \quad \Delta_2 = (x_2 - y_2) / 2.$$

By assumption $\Delta_1 \geq 0$ and $\Delta_2 \leq 0$. Furthermore, if $a \in (0, 1)$, then for a set of positive measure, $\Delta_1 > 0$ and $\Delta_2 < 0$, by our assumption that not all matching couples are identical.

Taking the average of the parents gives

$$(x_1x_2 + y_1y_2)/2 = x_1^*x_2^* + \Delta_1\Delta_2$$

and if $a \in (0, 1)$, this is strictly less than $x_1^*x_2^*$ for a set of positive measure. Plugging this into the integral gives

$$Cov(x_1, x_2) \leq Cov(x_1^*, x_2^*) = \int_K x_1^*x_2^* dk$$

with strict inequality if $a \in (0, 1)$. Since the parental covariance was 0 by assumption, this completes the proof.

□